Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		Motion of molecules remains unchanged (1) Collisions occur more often (1) Don't accept more collisions	2			2		2
		(ii)		It / volume decreases (1) to half **or** 9 cm^3 (1) Award 2 marks for inversely proportional so it halves		2		2	2	2
	(b)	(i)		300 [K]		1		1	1	1
		(ii)		Substitution: $\frac{2.4 \times 10^5}{300\ \textbf{ecf}} = \frac{3.0 \times 10^5}{T_2}$ (1) in any correct arrangement $T_2 = \frac{300\ \textbf{ecf} \times 3.0 \times 10^5}{2.4 \times 10^5}$ = 375 [K] (1) (= 102 °C) Temp rise = 375 – 300 **ecf** = 75 [K] or 75 °C (1) Alternative: $\frac{p_1}{p_2} = \frac{2.4 \times 10^5}{3.0 \times 10^5} = 0.8$ (1) $\frac{T_1}{T_2} = 0.8$ so $T_2 = \frac{300\ \textbf{ecf}}{0.8}$ = 375 [K] (1) Temp rise = 375 – 300 **ecf** = 75 [K] or 75 °C (1)	1	1 1		3	3	3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			Substitution into $Q = mc\Delta\theta$ to give $105.5 = 5 \times 10^{-3} \times c \times 21$ (1) Rearrangement: $c = \dfrac{105.5}{5 \times 10^{-3} \times 21}$ (1) = 1 004.8 [J/kg °C] (1) Accept 1 000 [J/kg °C] Don't accept 1 004 [J/kg °C] Award 2 marks for 1.005 or 1 [J/kg °C]	1 1	1		3	3	3
				Question 8 total	**4**	**7**	**0**	**11**	**9**	**11**